

Bellabeat Data-Driven Growth Strategy: Executive Insights & High-Level Recommendations

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Project: Google Data Analytics Capstone

Ask Phase: Business Task Summary

The primary objective is to analyze public smart device usage data from non-Bellabeat consumers to discover behavioral trends and patterns. These insights will guide the marketing analytics team in optimizing marketing strategies and product positioning—specifically for the **Bellabeat App**—to drive global growth in the smart device market.

Key Stakeholders

- **Urška Sršen:** Cofounder and Chief Creative Officer
- **Sando Mur:** Cofounder and executive team member
- **Bellabeat Marketing Analytics Team**

Initial Hypotheses

- **The Working Routine:** We expect weekday activity to match office hours, peaking during lunch and commuting times.
- **Weekend Shift:** We predict users will wake up later and have a more relaxed energy flow on Saturdays and Sundays.
- **Nightly Wind-Down:** We expect movement to drop significantly after 7:00 PM across the entire week.

Prepare Phase: Data Sources & Credibility

- **Data Source:** Fitbit Fitness Tracker Data (CC0: Public Domain, made available via Mobius on Kaggle).
- **Data Structure:** Personal tracking metrics captured by the second, minute, hour, and day. The core focus was narrowed down to daily_activity, hourly_steps, and hourly_intensities.
- **Data Integrity Check:** Using =COUNTUNIQUE in Google Sheets revealed a true distinct count of **33 unique users** (exceeding the initial baseline of 30).

ROCCC Assessment & Limitations Log

- **Reliable: Low/Medium.** It tracks a very small sample size of 33 individuals, which introduces statistical outlier risks and may not accurately generalize to the global market.
- **Original: Low.** Third-party data compiled from anonymized Fitbit submissions.
- **Comprehensive: Medium.** While it covers high-resolution minute/hourly logs, it lacks crucial demographic attributes (age, gender, location, occupation) necessary to confirm if these users fit Bellabeat's target audience.
- **Current: Low.** The data baseline is from 2016 (10 years old). Though core human biology stays the same, exact digital features and daily screen/wellness habits have evolved.
- **Cited: High.** Open-source documentation confirms a CC0: Public Domain classification with no privacy or copyright restrictions.

Scoping Strategy

The dataset was split into two separate monthly period folders: 12.03.16 - 11.04.16 and 12.04.16 - 12.05.16. Sorting and date-filtering exposed major sample drops (down to 18 users in some tables) and overlapping timelines within the first folder. Therefore, the analysis explicitly isolated the **12.04.16 - 12.05.16 folder**, providing a clean, consistent 33-user dataset covering a full uniform month. Hourly tables were chosen because they effectively summarized minute-level data while remaining clean enough to highlight macro trends.

Process Phase: Data Cleaning & Transformation Log

Data cleaning and manipulation were fully documented and scaled using **Google BigQuery** to ensure accuracy, formatting consistency, and structural integrity across all tables.

Daily Activity Table Pipeline

Cleaning Objectives & Data Rules

- **Irrelevant Data:** Excluded non-essential tracking metrics to focus strictly on core engagement variables.
- **Data Transformation:** Converted Id from an INTEGER data type to a STRING and trimmed any accidental leading/trailing whitespaces.
- **Filtering Logic:** Discarded entries containing 0 total steps to filter out non-wear time intervals.
- **Sorting:** Ordered the dataset by ActivityDate in descending order to display the most recent timelines first.

Step 1: Table Creation & Transformation

```
CREATE OR REPLACE TABLE
`skilful-answer-498915-k5.user_data.cleaned_daily_activity` AS
SELECT
  TRIM(CAST(Id AS STRING)) AS Id, -- Changes Id from INTEGER to STRING and trims
  whitespaces
  ActivityDate,
  TotalSteps -- Left out other columns/irrelevant data
FROM `skilful-answer-498915-k5.user_data.daily_activity`
WHERE TotalSteps > 0 -- Got rid of 0 step rows
ORDER BY ActivityDate DESC; -- Most recent date at the top
```

Step 2: Quality Assurance & Validation Checks

- NULL / Blank Cell Validation:

```
-- Validating that no null values or blank string errors exist
SELECT *
FROM `skilful-answer-498915-k5.user_data.cleaned_daily_activity`
WHERE Id IS NULL
OR ActivityDate IS NULL
OR TotalSteps IS NULL
OR CAST(TotalSteps AS STRING) = '';
```

- Duplicate Entry Validation:

```
-- Checking for duplicate rows based on unique composite keys (Id + ActivityDate)
SELECT
Id,
ActivityDate,
COUNT(*) AS row_count -- This creates a column to display what we want
FROM `skilful-answer-498915-k5.user_data.cleaned_daily_activity`
GROUP BY Id, ActivityDate
HAVING COUNT(*) > 1; -- This will show rows that are duplicated (exist greater
than 1 time)
```

- Distinct User Volume Count:

```
-- Verifying that the data captures the exact participant population sample
SELECT
COUNT(DISTINCT Id) AS users_count
FROM `skilful-answer-498915-k5.user_data.cleaned_daily_activity`; -- Returns 33
users
```

Hourly Intensities Table Pipeline

Cleaning Objectives & Data Rules

- **Irrelevant Data:** Removed unneeded metrics to optimize computing performance and maintain analytical clarity.
- **Data Transformation:** Standardized the Id column by casting it from INTEGER to STRING and applying a trimming function.
- **Sorting:** Ordered by ActivityHour in descending order to keep recent operational metrics accessible.

Step 1: Table Creation & Transformation

```
CREATE OR REPLACE TABLE
`skilful-answer-498915-k5.user_data.cleaned_hourly_intensities` AS
SELECT
  TRIM(CAST(Id AS STRING)) as Id, -- Got rid of whitespaces and converted Id
  (Integer) to String
  ActivityHour,
  TotalIntensity -- Only mentioned relevant columns / left out irrelevant columns
FROM `skilful-answer-498915-k5.user_data.hourly_intensities`
ORDER BY ActivityHour DESC;
```

Step 2: Quality Assurance & Validation Checks

- **NULL / Blank Cell Validation:**

```
-- Validating that no null values or blank string errors exist
SELECT *
FROM `skilful-answer-498915-k5.user_data.cleaned_hourly_intensities`
WHERE
  Id IS NOT NULL
  AND ActivityHour IS NOT NULL
  AND TotalIntensity IS NOT NULL
  AND activity_level IS NOT NULL;
```

- **Duplicate Entry Validation:**

```
-- Ensuring unique single hourly logs per user to prevent tracking skewness
SELECT
  Id,
  ActivityHour,
  COUNT(*) AS row_count -- This creates a column to display what we want
FROM `skilful-answer-498915-k5.user_data.cleaned_daily_intesities`
GROUP BY Id, ActivityHour
HAVING COUNT(*) > 1; -- This will show rows that are duplicated (exist greater
than 1 time)
```

- **Distinct User Volume Count:**

```
-- Ensuring full cohort visibility remains intact post-transformation
SELECT
  COUNT(DISTINCT Id) AS users_count
FROM `skilful-answer-498915-k5.user_data.cleaned_hourly_intensities`; -- Returns
33 users
```

Hourly Steps Table Pipeline

Cleaning Objectives & Data Rules

- **Irrelevant Data:** No columns required deletion from the source schema layout.
- **Data Transformation:** Cast the structural tracking IDs to string formatting and eliminated blank margins via trimming.
- **Sorting:** Ordered chronologically by ActivityHour in descending order.

Step 1: Table Creation & Transformation

```
-- TRIMMED whitespaces of STRINGS/ converted Id (INTEGER) to (STRING), Sorted
ActivityHour to DESC/ kept irrelevant columns out
CREATE OR REPLACE TABLE `skilful-answer-498915-k5.user_data.cleaned_hourly_steps`
AS
SELECT
  TRIM(CAST(Id AS STRING)) AS Id,
  ActivityHour,
  StepTotal
FROM `skilful-answer-498915-k5.user_data.hourly_steps`
ORDER BY ActivityHour DESC;
```

Step 2: Quality Assurance & Validation Checks

- **Duplicate Entry Validation:**

```
-- Confirming structural uniqueness constraints for active stepping logs
SELECT
  Id,
  ActivityHour,
  COUNT(*) AS row_count -- This creates a column to display what we want
FROM `skilful-answer-498915-k5.user_data.cleaned_hourly_steps`
GROUP BY Id, ActivityHour
HAVING COUNT(*) > 1; -- This will show rows that are duplicated (exist greater
than 1 time)
```

- **Distinct User Volume Count:**

```
-- Cross-checking base sample consistency across all hourly streaming sets
SELECT
  COUNT(DISTINCT Id) AS users_count
FROM `skilful-answer-498915-k5.user_data.cleaned_hourly_steps`; -- Returns 33
users
```

Analyze & Share Phase: Core Findings Summary

To uncover deep engagement trends, the separate datasets were joined and engineered using BigQuery so that Tableau could accurately read the dimensions and generate comprehensive heatmaps.

Step 1: Consolidation of Hourly Metrics Via LEFT JOIN

Found that the cleaned_hourly_intensities table (containing Id, ActivityHour, TotalIntensity, activity_level) and the cleaned_hourly_steps table (containing Id, ActivityHour, StepTotal) could be seamlessly joined because they shared matching composite keys (Id and ActivityHour). A LEFT JOIN was utilized to combine the tracking logs into a centralized dashboard-ready view:

```
-- CREATE OR REPLACE TABLE `skilful-answer-498915-k5.user_data.hourly_activity` AS
SELECT
  intensity.Id,
  intensity.ActivityHour,
  intensity.TotalIntensity,
  intensity.Activity_Level,
  steps.StepTotal -- This is the column I want to import from the hourly steps table
FROM `skilful-answer-498915-k5.user_data.cleaned_hourly_intensities` AS intensity
LEFT JOIN `skilful-answer-498915-k5.user_data.cleaned_hourly_steps` AS steps
  ON intensity.Id = steps.Id
  AND intensity.ActivityHour = steps.ActivityHour;
```


Step 2: Date Parsing & Temporal Feature Extraction

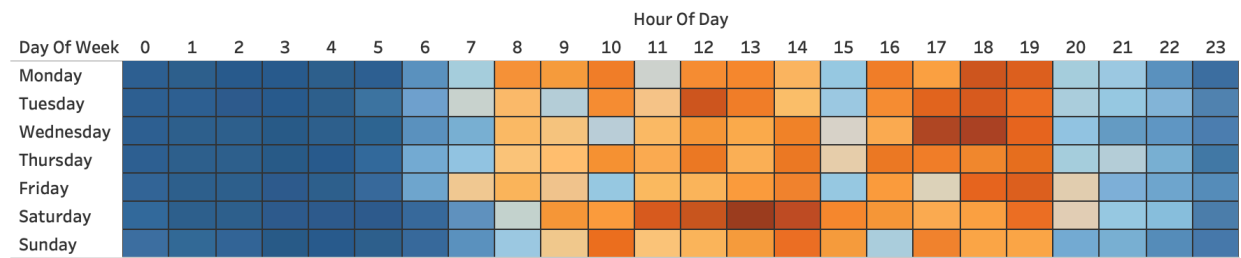
The following query extracts HourOfDay and the explicit day of the week from the combined ActivityHour timestamp, creating a new numeric HourOfDay column alongside an easily readable text-string DayOfWeek column. This ensures optimal parsing formatting for visual dashboard matrices:

```
-- Query utilized to isolate temporal intervals from the hourly activity view
SELECT
  Id,
  ActivityHour,
  -- 1. Extracts just the hour number (0 to 23)
  EXTRACT(HOUR FROM ActivityHour) AS HourOfDay,
  -- 2. Extracts the day of the week as a full text name (e.g., 'Sunday')
  FORMAT_TIMESTAMP('%A', ActivityHour) AS DayOfWeek,
  TotalIntensity,
  StepTotal
FROM `skilful-answer-498915-k5.user_data.hourly_activity`;
```

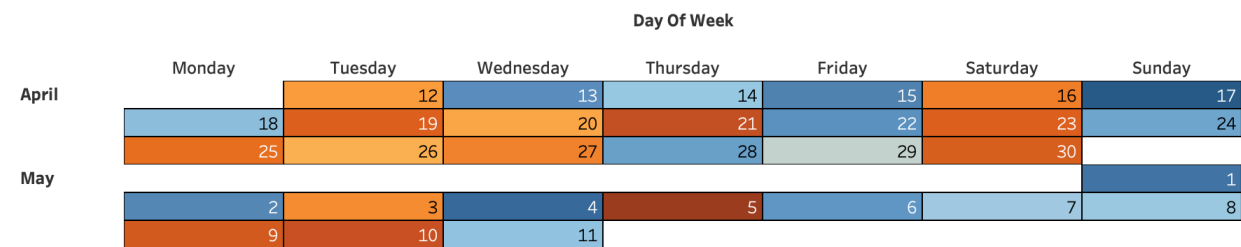
The same structural transformation had to be executed for the daily activity matrix to evaluate broader day-of-the-week lifestyle habits across the entire population cohort:

```
-- Query utilized to isolate explicit days of the week for macro step tracking
SELECT
  Id,
  ActivityDate,
  FORMAT_DATE('%A', ActivityDate) AS DayOfWeek,
  TotalSteps
FROM `skilful-answer-498915-k5.user_data.cleaned_daily_activity`;
```

Following the execution of these data transformation and feature engineering steps, the refined datasets were imported into Tableau to construct visual matrices. The resulting heatmaps below illustrate the exact intensity and step distribution patterns across different hours and days of the week, validating our initial hypotheses regarding user habits:



April - May 2016



Hourly Trends & Consumer Archetypes

- **The Weekday Routine (Mon–Fri):** Physical activity is heavily bound by a standard working routine. Sizable, consistent movement occurs from **8h to 19h**, with sharp spikes at midday lunch breaks (**12h–14h**) and post-work evening windows (**16h–19h**).
- **The Weekend Shift (Sat–Sun):** Users exhibit a delayed morning startup window, with movement picking up significantly later (**9h+**) than during the workweek (**7h/8h**). Saturdays feature a concentrated surge in activity from **11h to 14h**.
- **The Nightly Wind-Down:** Across all seven days, a sharp, uniform drop in movement begins **past 19h**, transitioning to an inactivity valley by **23h** as users sleep.

Daily Patterns & Behavioral Psychology

- **Peak Activity Days: Tuesdays** record the absolute highest average step volume of the week, followed closely by **Saturdays** as a secondary leisure-driven surge.
- **Low Activity Days: Sundays** mark a universal rest and recovery phase, bringing a massive drop to the lowest step counts of the week. **Fridays** serve as the lowest active window of the standard workweek due to accumulated fatigue.

Act Phase: Strategic Marketing & Product

Recommendations

By converting these behavioral trends into product features for the **Bellabeat App**, the company can position its ecosystem as an indispensable wellness companion.

Bellabeat Digital App Ecosystem	
Product Architecture	Marketing Campaign
1. Intelligent Notification Sync	11:45h "Midday Spark" Nudge (Before lunch spike) 16:00h "Post-Work Energy" Prompt (Before evening commute)
2. Sunset Wind-Down Mode	Post-19:00h Content Shift: Hydration, Sleep Prep & Blogs
3. Sunday Wellness Recap	Focus on Cumulative Metrics to Drive Brand Loyalty
4. Gamified E-Commerce Revenue Loop	Streak Points (Tue/Sat) -> Discounts on Hardware (Ivy/Leaf)

1. Implement an Intelligent Notification Sync Engine

- **Insight:** Users maximize their movement during highly predictable windows around their work schedules (commutes and lunch breaks).
- **Action:** Program the Bellabeat App to replace generic, rigid notifications with time-blocked, adaptive prompts. Deploy a **11:45h Midday Spark** nudge to encourage active lunch breaks, and a **16:00h Post-Work Energy** notification to capture the 16h–19h evening workout window. On weekends, automatically switch to a relaxed mode that silences early morning alerts to respect users' natural sleep cycles.

2. Launch the "Sunset Wind-Down" Content Campaign

- **Insight:** Activity consistently plunges past 19h across the entire week, revealing a natural boundary where users shift focus away from high-intensity physical movement.
- **Action:** Market Bellabeat as a holistic, around-the-clock wellness companion rather than just a fitness tracker. Past 19h, shift the app interface to focus on mental well-being and recovery. Use this window to automatically deliver guided meditations, hydration reminders, sleep-prep content, and curated knowledge hub articles (such as healthy recipes or expert wellness blogs).

3. Transition to a Sunday "Weekly Wellness Recap"

- **Insight:** Sundays show a universal, significant drop in physical movement, representing a natural rest day for users.
- **Action:** Stop pushing negative "missing your daily step goal" alerts on low-activity Sundays. Instead, deliver a celebrated **Weekly Wellness Recap** on Sunday evenings. Framing progress through cumulative weekly achievements rather than isolated daily goals reduces user frustration, builds long-term brand affinity, and fosters a sense of accomplishment.

4. Close the Loop with a Gamified E-Commerce Revenue Engine

- **Insight:** Weekly motivation routinely peaks on Tuesdays and Saturdays, showing clear windows of high user engagement.
- **Action:** Introduce a loyalty framework within the app where users earn **Bellabeat Wellness Points** for maintaining consistency streaks on high-motivation days (Tuesdays and Saturdays). Allow users to redeem these points directly for discounts on premium hardware (such as the Ivy, Leaf, or Time wellness watch) or subscription tiers. This transforms user data into an e-commerce lifecycle loop that drives hardware upsells and long-term retention.

Methodological Limitation Note

This analysis is built upon quantitative step and hourly logs from a public dataset. To fully refine these strategies, Bellabeat should pair these behavioral trends with qualitative consumer surveys to confirm target time zones and occupational demographics.